CLIPPING

A window is used to select that portion of the picture which is to be viewed. This is known as clipping. The process of clipping determines which elements of the picture lie inside the window and so are visible (i.e, displayed). The reverse of clipping is shielding - covering up a portion of the picture so that it is not visible.

Clipping is the process that determines the visible portion of the image lying inside the window. The points and lines that survive clipping are sent to the viewport by the window - to - viewport mapping.

def'n

Any procedure that identifies those portions of a picture that are either inside or outside of a specified region of space is referred to as a clipping algorithm or simply clipping

The region against which an object is to be clipped is called a clip window.

For the viewing transformations, we want to display only those picture parts that are within the window area. Everything outside the window is discarded. Clipping algorithms can be applied in the world coordinates, so that only the contents of the window interior are mapped to DC.

Different types of clipping are

→ Point clipping
→ Line clipping (straight line segment)
→ Area clipping (Polygon)
→ Curve clipping
→ Text clipping.

Aspect Ration

Since the viewing transformation involves scaling, undesirable distortion may be introduced whenever $S_x \neq S_y$. To avoid these distortions, the concept of aspect ration is needed.

The aspect ratio of a rectangular window or viewport is defined by :-

$$a = \frac{x_{max} - x_{min}}{y_{max} - y_{min}}$$

If the aspect ration a_w of the window equals the aspect ratio a_v of the viewport, then $S_x = S_y$, and no distortion (other than uniform magnification or compression) occurs. If $a_w \neq a_v$, then distortion occurs.

Rectangular Clipping Windows → we assume that our window is a rectangle whose sides are aligned with the coordinate axes. The x extent is measured from x_{min} to x_{max}, and the y extent is measured from y_{min} to y_{max}.

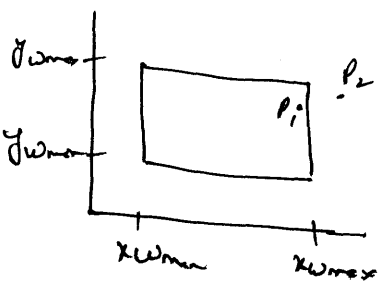
POINT CLIPPING →

A Point (x,y) is within the rectangular window if the x-coordinates between x_{min} and x_{max} and y-coordinate lies between y_{min} and y_{max}.

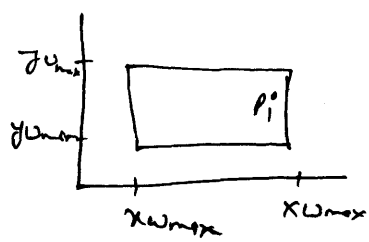
i.e a point at position (x, y) is saved from transformation to viewport if it satisfies the following inequalities :-

$$\left.\begin{aligned} xw_{min} \le x \le xw_{max} \\ yw_{min} \le y \le yw_{max} \end{aligned}\right] \quad -(1)$$

if any one of these 4 inequalities is not satisfied, the point is clipped



before clipping.



after clipping